

# Excercises Lab 6 Bonus

## DLP

### 1 Trivial Generators ( $g = 0$ or $g = 1$ )

In the Diffie-Hellman protocol, the security relies on the vast number of possible values for the public keys  $A = g^a \pmod{p}$  and the resulting shared secret.

- **Case  $g = 1$ :** Since  $1^k \equiv 1 \pmod{p}$  for any integer  $k$ , the public keys exchanged by both Alice and Bob will always be 1. Consequently, the shared secret will always be 1. An attacker can trivially predict the key.
- **Case  $g = 0$ :** Similarly,  $0^k \equiv 0 \pmod{p}$  (for  $k \geq 1$ ). The public keys and the shared secret will always be 0.

Both cases reduce the key space to a single value, rendering the encryption completely insecure.

### 2 Diffie-Hellman Calculation

Given the parameters  $(g, p) = (2, 197)$ , Joseph's private key  $a = 13$ , and Helen's private key  $b = 87$ .

The shared secret  $s$  is computed as:

$$s \equiv g^{ab} \pmod{p}$$

#### Step 1: Calculate the exponent

$$E = a \cdot b = 13 \cdot 87 = 1131$$

#### Step 2: Reduce exponent using Fermat's Little Theorem

Since  $p = 197$  is prime, we work modulo  $p - 1 = 196$ .

$$1131 \pmod{196}$$

$$1131 = 5 \times 196 + 151$$

Thus, we need to compute:

$$s \equiv 2^{151} \pmod{197}$$

### Step 3: Modular Exponentiation

Calculating  $2^{151} \pmod{197}$ :

$$2^7 = 128$$

$$2^8 = 256 \equiv 59$$

...

$$2^{151} \equiv 27 \pmod{197}$$

**Result:** The shared secret is 27.

### 3 Risks of Non-Generating $g$

If  $g$  is not a primitive root (generator) modulo  $p$ , it generates a subgroup of order  $q$ , where  $q$  is a divisor of  $p - 1$  and  $q < p - 1$ .

The values of  $g^x \pmod{p}$  will be confined to this smaller subgroup. This creates a vulnerability known as the **Small Subgroup Attack**.

- The search space for the private key is reduced from  $p - 1$  to  $q$ .
- If  $q$  is small, an attacker can solve the Discrete Logarithm Problem using brute force or algorithms like Pollard's Rho much faster than expected.

### 4 Security of Additive Groups

**Question:** Is Diffie-Hellman secure if the group operation is addition rather than multiplication?

**Answer:** No, it is not secure.

In a multiplicative group, security relies on the hardness of the Discrete Logarithm Problem: finding  $a$  given  $A = g^a \pmod{p}$ .

In an additive group, the public key generation becomes simple modular multiplication:

$$A = \underbrace{g + g + \cdots + g}_{a \text{ times}} \equiv a \cdot g \pmod{p}$$

To recover the private key  $a$ , an attacker only needs to solve for  $a$  in a linear congruence:

$$a \equiv A \cdot g^{-1} \pmod{p}$$

Since the modular multiplicative inverse  $g^{-1}$  can be computed efficiently using the **Extended Euclidean Algorithm** (in logarithmic time), the private key  $a$  can be recovered instantly.

## Example of Vulnerability

Let  $p = 17$ ,  $g = 3$ , and private key  $a = 5$ .

1. **Key Generation:**  $A = 5 \cdot 3 \equiv 15 \pmod{17}$ .

2. **Attack:** The attacker intercepts  $A = 15$ . They compute  $3^{-1} \pmod{17}$ .

$$3 \times 6 = 18 \equiv 1 \pmod{17} \implies g^{-1} = 6$$

3. **Recover Key:**  $a = 15 \cdot 6 = 90$ .

$$90 \pmod{17} = 5$$

The attacker successfully recovers  $a = 5$ .